MILITARY SPECIFICATION SHEET

# ELECTRON TUBE, TRANSMITTING

## TYPE 6000

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Beam power amplifier

| | | |
|---|---|---|
| Outline | | See figure 1 |
| Base | | |
| Envelope | --- | T11 |
| Cathode | --- | Coated unipotential |

Base connections:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| Pin No. | --- | 1 | 2 | 3 | 5 | 7 | 8 |
| Element | --- | g1 | h | k, g3 | a | h | g2 |

ABSOLUTE-MAXIMUM RATINGS:

| Parameter: | Ef | Eb | Ec1 | Ec2 | Ib | Ic1 | Pp | Pg2 | Pi(rf) | (A) Ehk | T(bulb) | F1 | (A) Alt |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Unit: | V | Vdc | Vdc | Vdc | mAdc | mAdc | W | W | W | v | °C | MHz | ft |
| Maximum: | 29.1 | 600 | -200 | 300 | 125 | 7 | 25 | 4.0 | 0.35 | 100 | 225 | 70 | (See |
| Minimum: | 23.8 | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | --- | note 1) |
| TEST CONDITIONS: | 26.5 | 250 | -18 | 250 | --- | --- | --- | --- | --- | --- | --- | --- | --- |

GENERAL:

Qualification - Required

(A) denotes changes



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| METHOD | REQUIREMENT OR TEST | CONDITIONS | AQL (PERCENT DEFECTIVE) | INSPECTION LEVEL OR CODE | SYMBOL | LIMITS MIN | LIMITS MAX | UNIT |
|---|---|---|---|---|---|---|---|---|
| | Quality conformance inspection, part 1 | | | | | | | |
| (A) 1266 | Total grid current | See note 2 | 1.0 | II | Ic | --- | -3.0 | μAdc |
| 1256 | Electrode current (anode) | | 1.0 | II | Ib | 50 | 90 | mAdc |
| 1256 | Electrode current (screen) | | 1.0 | II | Ic2 | --- | 10 | mAdc |
| 1236 | Power oscillation | Eb = 400 Vdc; Ec2 = 200 Vdc; Ib = 125 mAdc; Ic1 = 5.0 mAdc; Rg1 = 12,000 ohms; F = 70 MHz (see notes 2 and 3) | 1.0 | II | Po | 24 | --- | W |
| 1231 | Emission | Eb = Ec1 = Ec2 = 40 Vdc (see note 2) | 1.0 | II | Is | 300 | --- | mAdc |
| (A) 1201 | Short and discontinuity detection | | 0.4 | II | --- | --- | --- | --- |
| | Quality conformance inspection, part 2 | | | | | | | |
| 1301 | Heater current | | --- | --- | If | 250 | 310 | mA |
| 1336 | Heater-cathode leakage | | --- | --- | Ihk | --- | 75 | μAdc |
| 1306 | Transconductance | | --- | --- | Sm | 6,000 | 10,000 | μmhos |
| 1331 | Direct-interelectrode capacitance | No shield (see note 3) | --- | --- | Cgp | --- | 0.18 | pF |
| | | No shield (see note 3) | | | Cin | 11.5 | 18.5 | pF |
| | | No shield (see note 3) | | | Cout | 5.5 | 8.5 | pF |
| 1031 | Low-frequency vibration | Rp = 2,000 ohms | --- | --- | Ep | --- | 1,000 | mVac |
| (A) 1101 | Secureness of base, cap, or insert | | --- | --- | --- | --- | --- | --- |
| (A) 1111 | Base pin solder depth | | --- | --- | --- | --- | --- | --- |
| (A) 1211 | Insulation of electrodes | | 4.0 | S3 | --- | --- | --- | --- |
| (A) 1105 | Permanence of marking | | --- | --- | --- | --- | --- | --- |
| | Quality conformance inspection, part 3 | | | | | | | |
| 1501 | Intermittent life | Group A; Ec1 = -15 Vdc; Rg1 = 50,000 ohms | --- | --- | --- | --- | --- | --- |
| --- | Intermittent life-test end point (500 hours) | Power oscillation | --- | --- | Po | 20 | --- | W |



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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude maximum peak voltage.
2. This test to be performed at the conclusion of the holding period.
3. Base sleeve tied to ground.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - EL
Navy -
Air Force - 11, 70, 85
DSA - ES

User activities:
Army - AV
Navy - AS, OS, MC, CG, SH
Air Force - 19

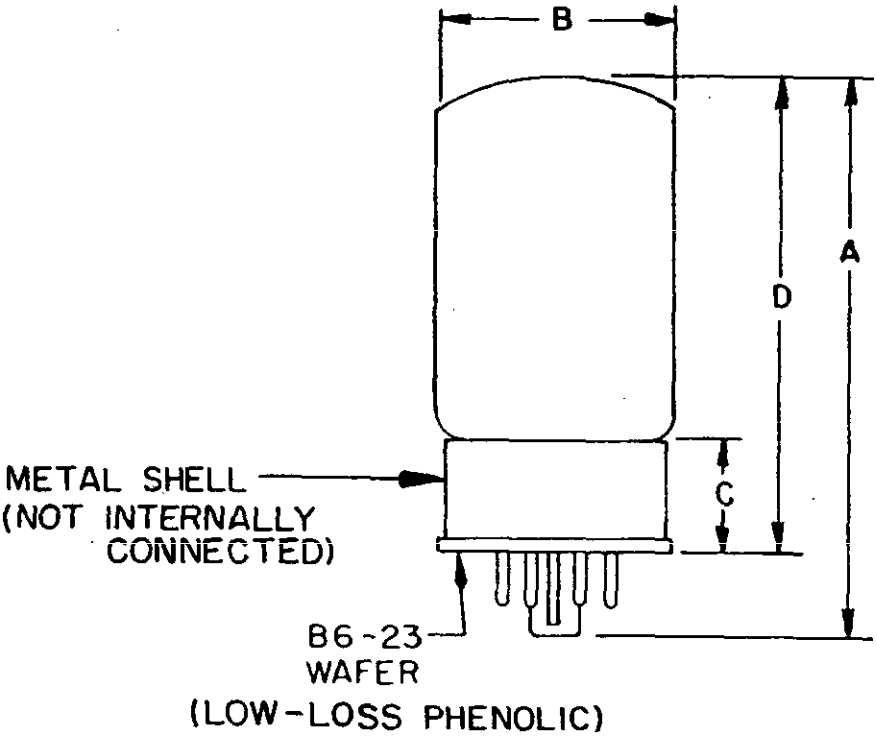
Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2401-9)



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| Ltr | Dimensions in inches with metric equivalents (mm) in parentheses |
|---|---|
| | *Nominal dimensions* |
| A | 3.47 (88.14) |
| B | 1.44 (36.58) |
| C | .640 (16.26) |
| D | 2.91 (73.91) |

FIGURE 1. Outline drawing of electron tube type 6000.



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